

## HW-4 Answers

a) Given

$$\Pr(X=1) = \theta_1$$

$$\Pr(X=2) = 2\theta_1$$

$$\Pr(X=3) = \theta_2$$

To be a valid Probability Mass Function (PMF), the total probability should add up to 1.

So, the total probability is,

$$\Pr(X=1) + \Pr(X=2) + \Pr(X=3) = \theta_1 + 2\theta_1 + \theta_2$$

$$\geq 3\theta_1 + \theta_2 = 1$$

Also, since these are probabilities.

$$\theta_1 \geq 0, \theta_2 \geq 0$$

So, the constraints should be,

$$i) 3\theta_1 + \theta_2 = 1$$

$$ii) \theta_1 \geq 0, \theta_2 \geq 0$$

b) Given

$s_1, s_2, s_3$  are the number of times values 1, 2, 3 occur in a dataset  $D$

$$D = \{x^{(1)}, x^{(2)}, \dots, x^{(n)}\}$$

Using the independent assumption,

$$\Pr(D|\theta) = \prod_{i=1}^n \Pr(x^{(i)}|\theta) = \theta_1^{s_1} (2\theta_1)^{s_2} \theta_2^{s_3}$$

$$= \theta_1^{s_1} \cdot 2^{s_2} \cdot \theta_1^{s_2} \cdot \theta_2^{s_3}$$

$$= 2^{s_2} \theta_1^{(s_1+s_2)} \theta_2^{s_3}$$

Applying log on both sides.

$$\log \Pr(D|\theta) = \log (2^{s_2} \theta_1^{(s_1+s_2)} \theta_2^{s_3})$$

$$= s_2 \log 2 + (s_1+s_2) \log \theta_1 + s_3 \log \theta_2$$

c) To find MLE of  $\theta$  based on sequence D.

We maximize the log-likelihood under the constraint  $3\theta_1 + \theta_2 = 1$

We know that,  $3\theta_1 + \theta_2 = 1$

$$\log \Pr(D|\theta) = s_2 \log 2 + (s_1 + s_2) \log \theta_1 + s_3 \log \theta_2$$

We apply the method of Lagrange multipliers,

$$\mathcal{L}(\theta_1, \theta_2, \lambda) = (s_1 + s_2) \log \theta_1 + s_3 \log \theta_2 + \lambda(1 - 3\theta_1 - \theta_2)$$

(We ignore  $s_2 \log 2$ , as it is a constant)

Set partial derivatives to 0.

$$\frac{d\mathcal{L}}{d\theta_1} = \frac{s_1 + s_2}{\theta_1} - 3\lambda = 0 \Rightarrow \frac{s_1 + s_2}{\theta_1} = 3\lambda \quad \text{--- (1)}$$

$$\frac{d\mathcal{L}}{d\theta_2} = \frac{s_3}{\theta_2} - \lambda = 0 \Rightarrow \frac{s_3}{\theta_2} = \lambda \quad \text{--- (2)}$$

Applying (2) in (1), we get,

$$\frac{s_1 + s_2}{\theta_1} = 3 \cdot \frac{s_3}{\theta_2} \Rightarrow \frac{s_1 + s_2}{\theta_1} = \frac{3s_3}{\theta_2}$$

$$\Rightarrow \frac{\theta_2}{\theta_1} = \frac{3s_3}{s_1 + s_2}$$

Substituting this in  $3\theta_1 + \theta_2 = 1$

$$\Rightarrow 3\theta_1 + \frac{3s_3}{s_1 + s_2} \theta_1 = 1 \Rightarrow \theta_1 = \frac{1}{3 + \frac{3s_3}{s_1 + s_2}}$$

$$\Rightarrow \hat{\theta}_1 = \frac{s_1 + s_2}{3(s_1 + s_2 + s_3)}$$

$$\text{Then, } \theta_2 = 1 - 3\theta_1 = 1 - \frac{3(s_1 + s_2)}{3(s_1 + s_2 + s_3)}$$

$$\Rightarrow \hat{\theta}_2 = \frac{s_3}{s_1 + s_2 + s_3}$$

2) Given,  $\{x^{(1)}, \dots, x^{(n)}\}$  is the iid sample from an exponential distribution,

Density function,  $f(x|\beta) = \frac{1}{\beta} \exp\left(-\frac{x}{\beta}\right)$  for  $0 \leq x < \infty$



To find,

MLE of the parameter  $\beta$ .

The likelihood function,

$$L(\beta) = \prod_{i=1}^n \frac{1}{\beta} e^{-x^{(i)}/\beta}$$

$$= \left(\frac{1}{\beta}\right)^n \exp\left(-\frac{\sum x^{(i)}}{\beta}\right)$$

Log-likelihood:

$$\log L(\beta) = \log\left(\left(\frac{1}{\beta}\right)^n \exp\left(-\frac{\sum x^{(i)}}{\beta}\right)\right)$$

$$= -n \log \beta - \frac{1}{\beta} \sum_{i=1}^n x^{(i)}$$

Take derivative & set to zero.

$$\frac{d}{d\beta} \log L(\beta) = -\frac{n}{\beta} + \frac{1}{\beta^2} \sum x^{(i)} = 0$$

Multiply both sides by  $\beta^2$ ,

$$\Rightarrow -n\beta + \sum x^{(i)} = 0$$

$$\Rightarrow \beta = \frac{1}{n} \sum x^{(i)} = \bar{x}$$

Thus, MLE of the parameter  $\beta$  is,

$$\hat{\beta} = \frac{1}{n} \sum_{i=1}^n x^{(i)} = \bar{x}$$

————— X —————

3a) Given,

We observe 5 defective out of 30.

Prior on  $\theta$  is uniform on  $[0, 1]$ .

To find,

Posterior up to a constant.

We know that,

Here, we have a Binomial likelihood with a Beta prior:

$$\Rightarrow \text{Likelihood: } \Pr(D|\theta) \propto \theta^5(1-\theta)^{25}$$

$$\Rightarrow \text{Prior: } p(\theta) = 1 \text{ (uniform on } [0,1])$$

$$\Rightarrow \text{Posterior: } p(\theta|D) \propto \theta^5(1-\theta)^{25}$$

As Posterior = Likelihood  $\times$  Prior

And the Constant term in likelihood is dropped.

$$\text{So, Posterior } p(\theta|D) \propto \theta^5(1-\theta)^{25}$$

This is a Beta(6, 26) distribution //

3) b) Given:

~~Likelihood:  $\sim \mathcal{N}$~~

3) b) Given,

$$\text{Likelihood} \sim \mathcal{N}(\mu, 1)$$

$$\text{Prior} \sim \mathcal{N}(0, 1)$$

$$\text{Distribution } D = \{x^{(1)}, \dots, x^{(n)}\}$$

To find,

Posterior distribution of  $\mu$ .

We know that, as posterior is also Normal,

$$\text{Posterior mean, } \mu_n = \frac{n\bar{x}}{n+1}$$

Posterior variance,

$$\sigma_n^2 = \frac{1}{n+1}$$

So,

$$p(\mu|D) = \mathcal{N}\left(\frac{n\bar{x}}{n+1}, \frac{1}{n+1}\right) //$$